Theory of Elementary Particles: Lecture 6

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Tensor product
SU(2)

Graphical representation of SU(2)

SU(3) group

Graphical representation of SU(3)

SU(4)

Tensor method and $SU(n)$

SU(n): tensor approach

Tensors and permutation symmetry

Young Tableaux

Definition of tensor product

Tensor product $V \otimes U$ of two linear spaces V and U over real or complex number is the Cartesian product of both spaces modulo the below identifications of elements, namely

$$(\lambda e_j) \otimes f_k \equiv e_j \otimes (\lambda f_k) = \lambda(e_j \otimes f_k) \quad \forall e_j \in V, f_k \in U, \quad \lambda \in \mathbb{R} \,\mathrm{or}\, \mathbb{C} \tag{1}$$

We define

$$D(a)(u \otimes v) \equiv D(a)u \otimes D(a)v \tag{2}$$

We may think about $D(a)$ as a large matrix built from the two representations acting in the spaces U and V

Lie algebra

if $M = \exp(i\epsilon X) \approx 1 + i\epsilon X$ then

$$M(a \otimes b) = Ma \otimes Mb = (1 + i\epsilon X)a \otimes (1 + i\epsilon X)b = a \otimes b + i\epsilon(Xa \otimes b + a \otimes Xb) \quad (3)$$

so we see that the natural action of the elements of $SU(n)$ Lie algebra on the tensor product of two representation (1) and (2) is so that the generators are defined as

$$\mathbf{J} = \mathbf{J}^{(1)} + \mathbf{J}^{(2)}, \quad \mathbf{J}^{(1)} \equiv \mathbf{J}^{(1)} \otimes \mathbf{1}, \quad \mathbf{J}^{(2)} \equiv \mathbf{1} \otimes \mathbf{J}^{(2)} \quad (4)$$

- $$r'_i = U_{ij}(\boldsymbol{\epsilon}) r_j, \quad s'_i = U_{ij}(\boldsymbol{\epsilon}) s_j$$
- Tensor product

$$r'_i s'_k = U_{ij}(\boldsymbol{\epsilon}) U_{kl}(\boldsymbol{\epsilon}) r_j s_l = D_{ik,jl}(\boldsymbol{\epsilon})(r_j s_l)$$

-
$$\begin{aligned} r'_j &\approx (1 + i\epsilon_a J_a^{(1)})_{jl} r_l \\ s'_j &\approx (1 + i\epsilon_a J_a^{(2)})_{jl} s_l \end{aligned}$$

$$r'_i s'_k \approx \left(1 + i\epsilon_a (J_a^{(1)})_{il}\delta_{nk} + i\epsilon_a (J_a^{(2)})_{kn}\delta_{li}\right) r_l s_n \tag{5}$$

- Tensor product of j_1 and j_2 spin representations, let us find the irreducible representations.
- We construct Casimir operator

$$\begin{aligned}
\mathbf{J}^2 &= (\mathbf{J}^{(1)} + \mathbf{J}^{(2)})^2 && (6)\\
&= (\mathbf{J}^{(1)})^2 + (\mathbf{J}^{(2)})^2 + 2\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)} && (7)\\
&= (\mathbf{J}^{(1)})^2 + (\mathbf{J}^{(2)})^2 + 2\left[\frac{1}{2}\left(J_+^{(1)} J_-^{(2)} + J_-^{(1)} J_+^{(2)}\right) + J_3^{(1)} J_3^{(2)}\right], && (8)
\end{aligned}$$

where

$$J_\pm^{(i)} = J_1^{(i)} \pm i J_2^{(i)} \qquad (9)$$

Then

$$\begin{aligned}
J_+^{(1)} J_-^{(2)} &= J_1^{(1)} J_1^{(2)} - i J_1^{(1)} J_2^{(2)} + i J_2^{(1)} J_1^{(2)} + J_2^{(1)} J_2^{(2)} && (10)\\
J_-^{(1)} J_+^{(2)} &= J_1^{(1)} J_1^{(2)} + i J_1^{(1)} J_2^{(2)} - i J_2^{(1)} J_1^{(2)} + J_2^{(1)} J_2^{(2)} && (11)
\end{aligned}$$

Raising/lowering operators and the J_3 operator

$$J_\pm = J_\pm^{(1)} + J_\pm^{(2)} \tag{12}$$

$$J_3 = J_3^{(1)} + J_3^{(2)} \tag{13}$$

Example

- Consider the tensor product:

$$\frac{1}{2} \otimes \frac{1}{2} = ? \tag{14}$$

- the basis which spans the representation must be of the form

$$| \frac{1}{2}, \frac{a}{2} \rangle_{(1)} | \frac{1}{2}, \frac{b}{2} \rangle_{(2)}, \quad a, b = \pm 1$$

- the highest J_3 eigenvalue state reads

$$| \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)}$$

indeed

$$\begin{aligned}
* &= J_3 | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} && (15)\\
&= (J_3^{(1)} \otimes 1) | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} + (1 \otimes J_3^{(2)}) | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} && (16)\\
&= \frac{1}{2} | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} + \frac{1}{2} | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} && (17)\\
&= 1 | \frac{1}{2}, \frac{1}{2} \rangle_{(1)} | \frac{1}{2}, \frac{1}{2} \rangle_{(2)} && (18)
\end{aligned}$$

- hence, it must belong to the spin-1 representation, to check calculate the Casimir operator eigenvalue

- $$\mathbf{J}^2 \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} = \left(\frac{3}{2}+\frac{1}{2}\right) \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} = 2 \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \tag{19}$$

  so $2 = j(j+1)$ and $j = 1$
- hence

$$\mid 1,1\rangle \equiv \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \tag{20}$$

Cross check:

$$* = \left\{ (\mathbf{J}^{(1)})^2 + (\mathbf{J}^{(2)})^2 + 2\left[\frac{1}{2}\left(J_+^{(1)}J_-^{(2)} + J_-^{(1)}J_+^{(2)}\right) + J_3^{(1)}J_3^{(2)}\right]\right\} \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \tag{21}$$

$$= \left\{ (\mathbf{J}^{(1)})^2 + (\mathbf{J}^{(2)})^2 + 2J_3^{(1)}J_3^{(2)}\right\} \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \tag{22}$$

$$= \left\{ \frac{3}{4}+\frac{3}{4}+2\frac{1}{2}\frac{1}{2}\right\} \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \tag{23}$$

-

$$\sqrt{2} \mid 1,0\rangle = J_- \mid 1,1\rangle \quad (24)$$

$$= (J_-^{(1)} + J_-^{(2)}) \mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \quad (25)$$

$$= \left(\mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},-\frac{1}{2}\rangle_{(2)} + \mid \frac{1}{2},-\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \right) \quad (26)$$

- The orthogonal state to $\mid 1,0\rangle$ is

$$\mid *\rangle = \frac{1}{\sqrt{2}} \left(\mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},-\frac{1}{2}\rangle_{(2)} - \mid \frac{1}{2},-\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \right) \quad (27)$$

Notice that

$$\mathbf{J}^2 \mid *\rangle = \left(\frac{3}{2} - 1 - \frac{2}{4} \right) \mid *\rangle = 0 \quad (28)$$

so it is spin 0 state!so $\mid *\rangle = \mid 0,0\rangle$

- the lowest-lying state of spin 1 reads

$$\sqrt{2} \mid 1,-1\rangle = J_- \mid 1,0\rangle \quad (29)$$

$$= (J_-^{(1)} + J_-^{(2)}) \frac{1}{\sqrt{2}} \left(\mid \frac{1}{2},\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},-\frac{1}{2}\rangle_{(2)} + \mid \frac{1}{2},-\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},\frac{1}{2}\rangle_{(2)} \right)$$

$$= \sqrt{2} \mid \frac{1}{2},-\frac{1}{2}\rangle_{(1)} \mid \frac{1}{2},-\frac{1}{2}\rangle_{(2)} \quad (30)$$

Summary:

- $$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1 \tag{31}$$
- Spin-1 representation (triplet) which contains three states:

$$|\ 1,\ 1\rangle = |\ \frac{1}{2}, \frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, \frac{1}{2}\rangle_{(2)} \tag{32}$$

$$|\ 1,\ 0\rangle = \frac{1}{\sqrt{2}}\left(|\ \frac{1}{2}, \frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(2)} + |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, \frac{1}{2}\rangle_{(2)}\right) \tag{33}$$

$$|\ 1, -1\rangle = |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(2)} \tag{34}$$

  which are symmetric when one interchanges $(1) \leftrightarrow (2)$
- Spin-0 representation (singlet) which contains one state

$$|\ 0,\ 0\rangle = \frac{1}{\sqrt{2}}\left(|\ \frac{1}{2}, \frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(2)} - |\ \frac{1}{2}, -\frac{1}{2}\rangle_{(1)}\ |\ \frac{1}{2}, \frac{1}{2}\rangle_{(2)}\right) \tag{35}$$

  antisymmetric if one interchanges $(1) \leftrightarrow (2)$

The procedure of working out the irreducible representations of the product representations can be summarized as follows:

(i) Start with the combination of states with largest J_3. This is also the state with the largest total J.

(ii) Use the lowering operator $J_- = J_-^{(1)} + J_-^{(2)}$ to get to all the others states in the same irreducible representation.

(iii) Find the orthogonal combination to $|J_m, J_m - 1\rangle$ where J_m is the maximum value of J in the product. This should be the state $|\ J_m - 1, J_m - 1\rangle$. Then use the lowering operator to reach the other $J = J_m - 1$ states.

(iv) Repeat these steps until $J = |\ j_1 - j_2\ |$

Tensor product of two $SU(2)$ representations

The product representations $\mid j_1, m_1\rangle$ and $\mid j_2, m_2\rangle$ can be combined into eigenstates of $\mid J, M\rangle$ of total $\mathbf{J} = \mathbf{J^{(1)}} + \mathbf{J^{(2)}}$:

$$\mid J, M\rangle = \sum_{m_1, m_2} \langle j_1 m_1 j_2 m_2 \mid JM\rangle \mid j_1 m_1\rangle \mid j_2 m_2\rangle \tag{36}$$

where $\langle j_1 m_1 j_2 m_2 \mid JM\rangle$ is the Clebsch-Gordan coefficient.

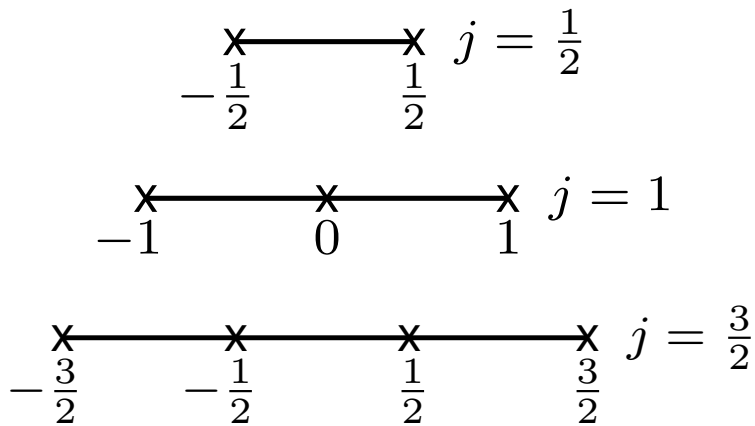
$$J = j_1 + j_2, j_1 + j_2 - 1, ..., \mid j_1 - j_2 \mid \tag{37}$$

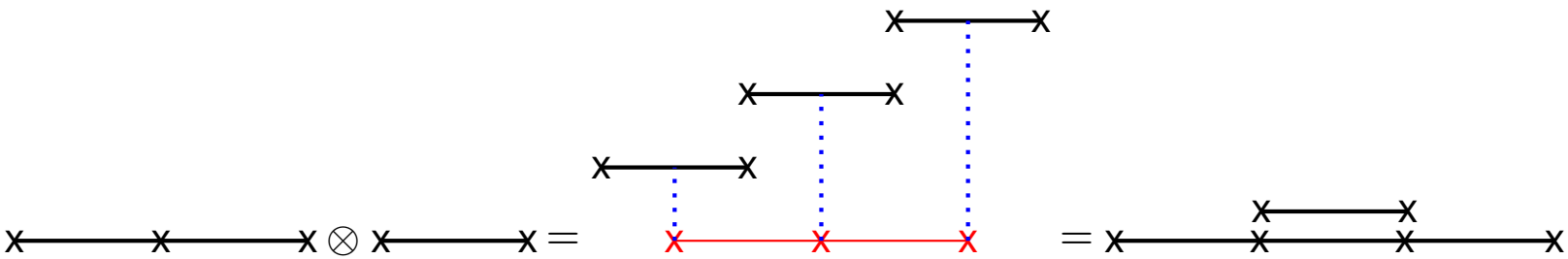
$$m_i = -j_i, -j_i + 1, ..., j_i - 1, j_i \tag{38}$$

Theorem ($\rightarrow$ exercise)

Tensor product of two representations of spin j and j' can be reduced to the sum of $N = 2\min(j, j') + 1$ irreducible representations of spin $j + j'$, $j + j' - 1$,..., $|j - j'|$.

Graphical representation of SU(2)





$$1 \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{3}{2} \tag{39}$$

SU(3) group

- $SU(3)$ group

$$U = \exp\left(\frac{i}{2}\sum_i \theta_i \lambda_i\right) \tag{40}$$

- $8 = 3^2 - 1$ generators: Gell-Mann matrices λ_is

$$\lambda_1 = \begin{pmatrix} 0 & 1 & \cdot \\ 1 & 0 & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & \cdot \\ i & 0 & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}, \quad \lambda_3 = \begin{pmatrix} 1 & 0 & \cdot \\ 0 & -1 & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix},$$

$$\lambda_4 = \begin{pmatrix} 0 & \cdot & 1 \\ \cdot & \cdot & \cdot \\ 1 & \cdot & 0 \end{pmatrix}, \quad \lambda_5 = \begin{pmatrix} 0 & \cdot & -i \\ \cdot & \cdot & \cdot \\ i & \cdot & 0 \end{pmatrix},$$

$$\lambda_6 = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & 0 & 1 \\ \cdot & 1 & 0 \end{pmatrix}, \quad \lambda_7 = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & 0 & -i \\ \cdot & i & 0 \end{pmatrix},$$

$$\lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

- The commutation relations:

$$\left[\frac{1}{2}\lambda_i, \frac{1}{2}\lambda_j\right] = \frac{i}{2} f_{ijk}\lambda_k, \tag{41}$$

where:

$$f_{123} = 1$$

$$f_{458} = f_{678} = \frac{\sqrt{3}}{2} \tag{42}$$

$$f_{147} = f_{246} = f_{257} = f_{345} = f_{516} = f_{637} = \frac{1}{2} \tag{43}$$

- Anticommutation relations:

$$\{\lambda_i, \lambda_j\} = \frac{4}{3}\delta_{ij}\,\mathbf{1} + 2d_{ijk}\lambda_k. \tag{44}$$

Let us introduce:

- generators F_is:

$$F_i = \frac{1}{2}\lambda_i \tag{45}$$

- Isospin operator:

$$T_3 = F_3 \tag{46}$$

- Hypercharge operator

$$Y = \frac{2}{\sqrt{3}}F_8 = \frac{2}{\sqrt{3}}\frac{1}{2}\lambda_8 \tag{47}$$

- raising/lowering operators

$$T_\pm = T_1 \pm iT_2 \tag{48}$$
$$U_\pm = F_6 \pm iF_7 \tag{49}$$
$$V_\pm = F_4 \pm iF_5 \tag{50}$$

Commutation relations (check $\rightarrow$ exercise)

$$
\begin{aligned}
[T_3, T_\pm] &= \pm T_\pm && (51)\\
[Y\ , T_\pm] &= 0 && (52)\\
[T_3, U_\pm] &= \mp\frac{1}{2} U_\pm && (53)\\
[Y\ , U_\pm] &= \pm U_\pm && (54)\\
[T_3, V_\pm] &= \pm\frac{1}{2} V_\pm && (55)\\
[Y\ , V_\pm] &= \pm V_\pm && (56)\\
[T_+, T_-] &= 2T_3 && (57)\\
[T_+, V_+] &= [T_+, U_-] && (58)\\
&= [U_+, V_+] = 0 && (59)\\
[T_+, V_-] &= -U_- && (60)\\
[T_+, U_+] &= V_+ && (61)\\
[U_+, V_-] &= T_- && (62)\\
[Y\ , T_3] &= 0 && (63)
\end{aligned}
$$

$$
\begin{aligned}
[U_+, U_-] &= \frac{3}{2} Y - T_3 \equiv 2U_3 && (64)\\
[V_+, V_-] &= \frac{3}{2} Y + T_3 \equiv 2V_3 && (65)
\end{aligned}
$$

$SU(3)$ contains three $SU(2)$ subgroups:

- isospin I generated by: T_1, T_2 and T_3;
- U-spin generated by: $U_\pm$, U_3;
- V-spin generated by: $V_\pm$, V_3.

- Fundamental representation is given by the triplet $q = (u, d, s)$
- Within triplet q there are three iso-doublets: $u, d,\ (T)$, $d, s,\ (U)$, $u, s,\ (V)$.
- Generator $\lambda_3/2$ is an isospin operator with eigenvalues $1/2$, $-1/2$, 0 for the states u, d, s.
- We may define the Casimir operator:

$$C = \mathbf{F}^2 = \sum_{i=1}^{8} F_i^2 = \frac{1}{2}\{T_+, T_-\} + T_3^2 + \frac{1}{2}\{U_+, U_-\} + \frac{1}{2}\{V_+, V_-\} + F_8^2$$

- $SU(3)$ has rank 2, indeed there are two diagonal operators

$$[T_3, Y] = 0$$

- Hence the state of given finite-dimensional representation can be labelled by eigenvalues of Y and T_3 or Q and T_3, where the charge operator Q is defined by

$$Q = T_3 + \frac{1}{2}Y \tag{66}$$

- Notice that

$$[U_3, Q] = \frac{3}{4}[Y, T_3] - \frac{1}{4}[T_3, Y] = [Y, T_3] = 0 \tag{67}$$

- Let us introduce the state $\mid y, t_3\rangle$ so that

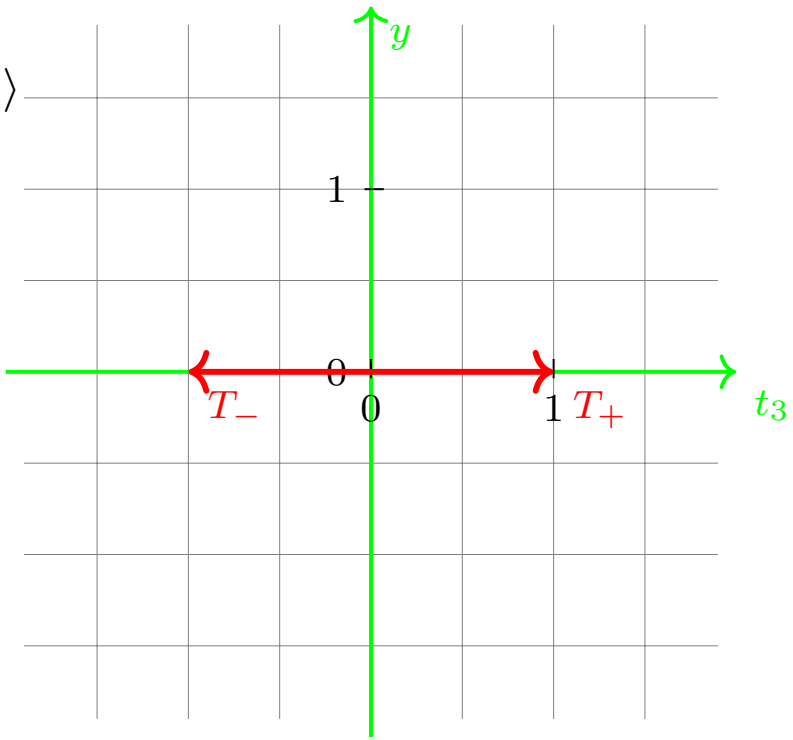
$$T_3 \mid y, t_3\rangle = t_3 \mid y, t_3\rangle, \quad Y \mid y, t_3\rangle = y \mid y, t_3\rangle$$

- We easily see that

$$\begin{aligned} T_3 T_\pm \mid y, t_3\rangle &= (\underbrace{[T_3, T_\pm]}_{\pm T_\pm} + T_\pm T_3) \mid y, t_3\rangle \\ &= (t_3 \pm 1) T_\pm \mid y, t_3\rangle \quad (68) \\ Y T_\pm \mid y, t\rangle &= (\underbrace{[Y, T_\pm]}_{0} + T_\pm Y) \mid y, t\rangle \quad (69) \\ &= y T_\pm \mid y, t\rangle \quad (70) \end{aligned}$$

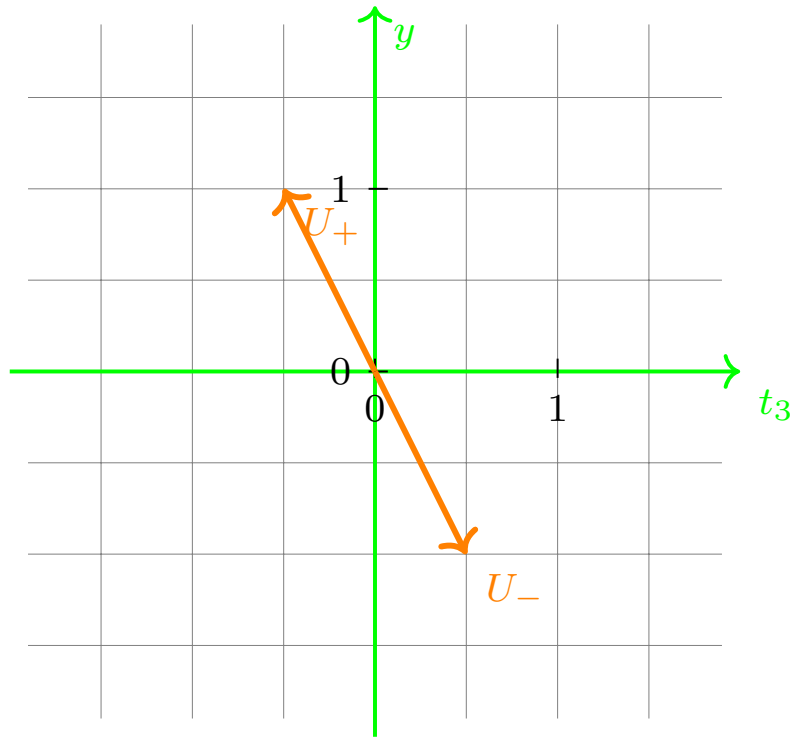
so

$$T_\pm \mid y, t\rangle \sim \mid y, t_3 \pm 1\rangle$$



- Similarly

$$
\begin{aligned}
T_3 U_\pm \mid y,t\rangle &= (\underbrace{[T_3,U_\pm]}_{\mp\frac{1}{2}U_\pm} + U_\pm T_3) \mid y,t\rangle \\
&= \left(t \mp \frac{1}{2}\right) U_\pm \mid y,t\rangle \quad (71) \\
Y U_\pm \mid y,t\rangle &= (\underbrace{[Y,U_\pm]}_{\pm U_\pm} + U_\pm Y) \mid y,t\rangle \\
&= (y \pm 1) U_\pm \mid y,t\rangle \quad (72)
\end{aligned}
$$

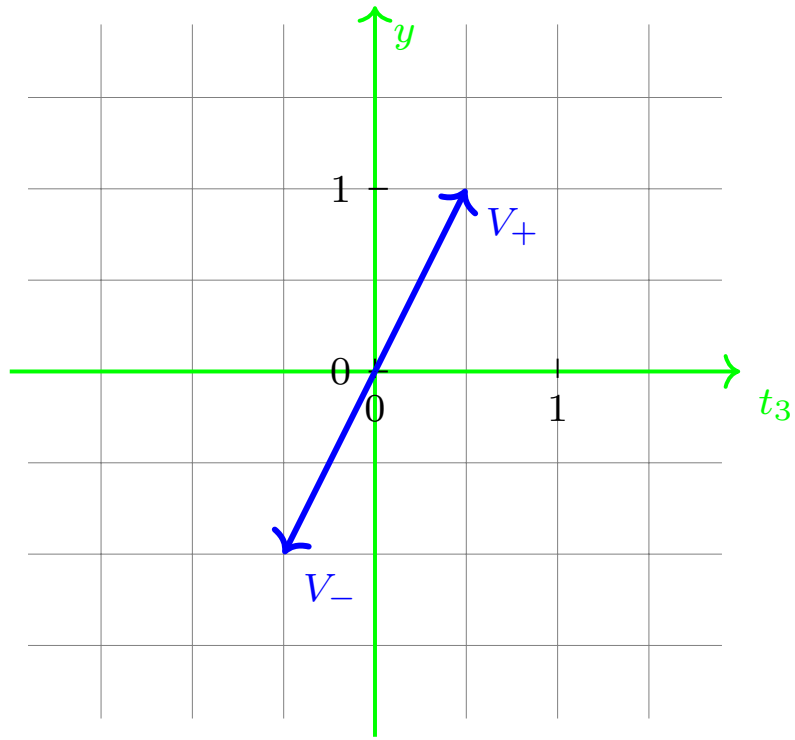


so

$$U_\pm \mid y, t_3\rangle \sim \mid y \pm 1, t_3 \mp \frac{1}{2}\rangle \quad (73)$$

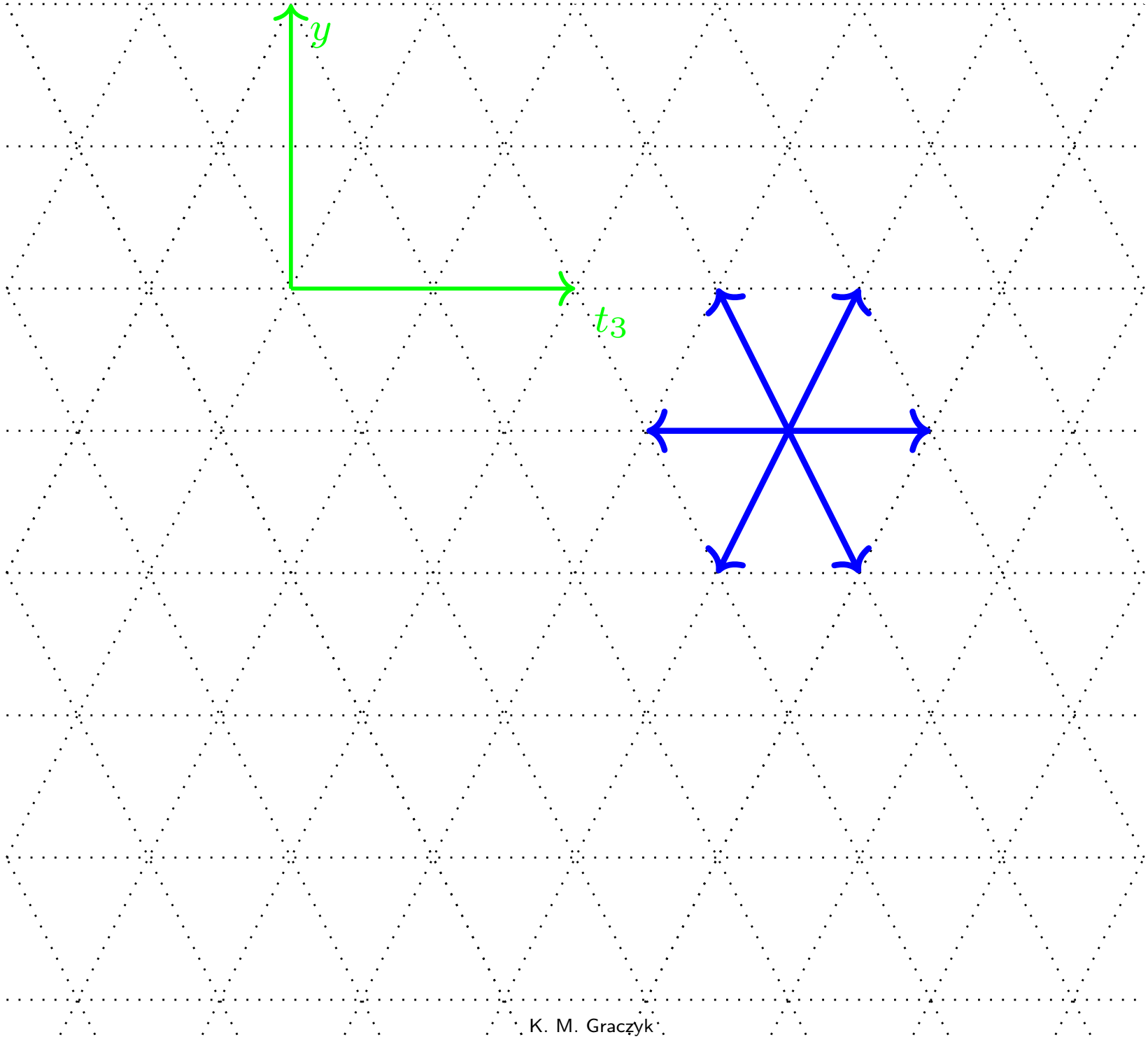
- Eventually

$$
\begin{aligned}
T_3 V_\pm \mid y,t\rangle &= (\underbrace{[T_3,V_\pm]}_{\pm\frac{1}{2}V_\pm} + V_\pm T_3) \mid y,t\rangle \\
&= \left(t \pm \frac{1}{2}\right) V_\pm \mid y,t\rangle \quad (74) \\
Y V_\pm \mid y,t\rangle &= (\underbrace{[Y,V_\pm]}_{\pm V_\pm} + V_\pm Y) \mid y,t\rangle \\
&= (y \pm 1) V_\pm \mid y,t\rangle \quad (75)
\end{aligned}
$$

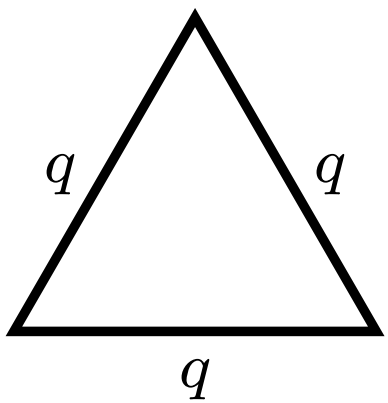


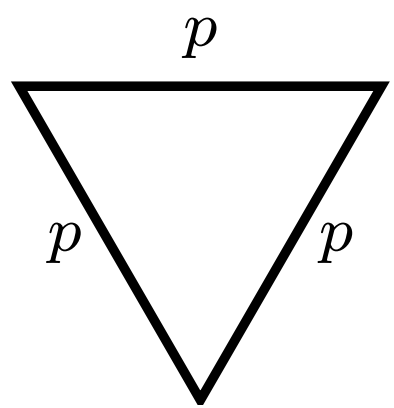
so

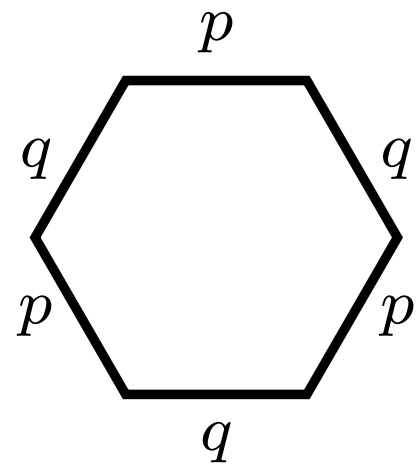
$$V_\pm \mid y, t_3\rangle \sim \mid y \pm 1, t_3 \pm \frac{1}{2}\rangle$$


y
t_3

- Each representation of $SU(3)$ is characterized by two integers (p, q)
- Graphically given representation can be represented by hexagonal figure
  - symmetric with respect to y-axis, because $SU(2)$ subgroup
  - three sides have p units of length, and three q units of length
  - the hexagon collapses into a equilateral triangle when either p or q vanishes







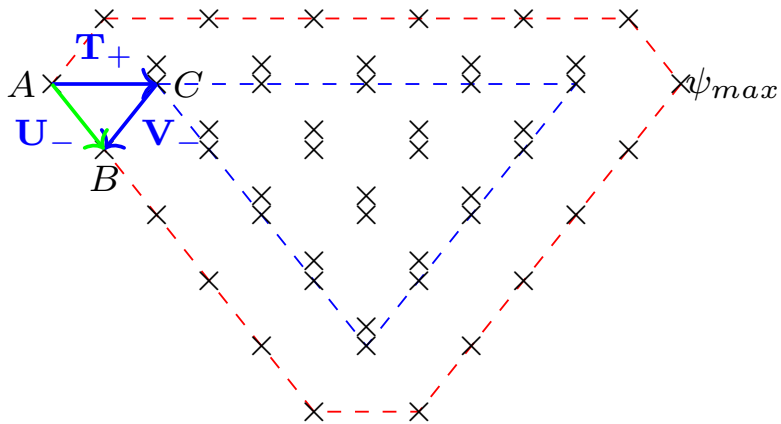
Graphical representation of SU(3)

- in order to reconstruct the full finite dimensional representation we must consider the state $\mid y_{max}, t_{3,max}\rangle$ and using $V_{\pm}$, $U_{\pm}$ and $T_{\pm}$ recover the full set of states.
- it is easy to see that there exists only one state on the boundary

$$0 = V_- \underbrace{T_+ \mid A\rangle}_{=0} \quad (76)$$

$$= (\underbrace{[V_-, T_+]}_{U_-} + T_+ V_-) \mid A\rangle$$

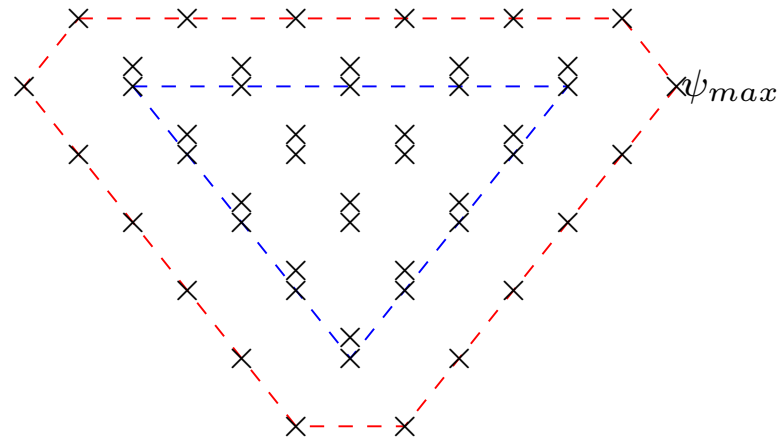
$$\Rightarrow U_- \mid A\rangle = -T_+ V_- \mid A\rangle \equiv \mid B\rangle \quad (77)$$



the sites in:

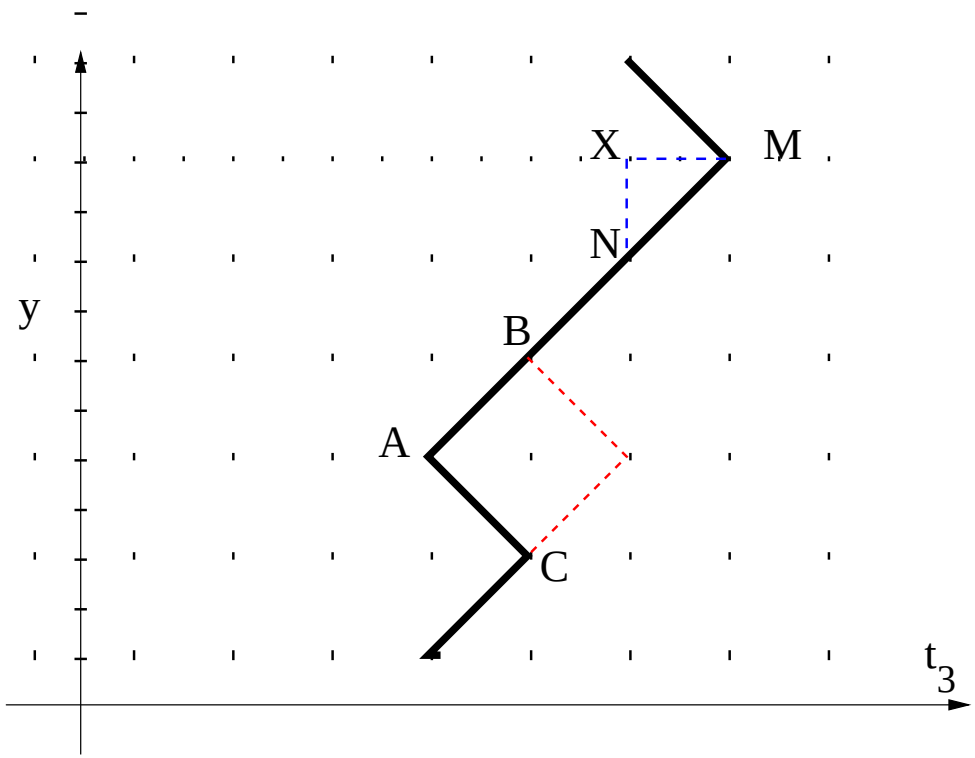
- the boundary: singly occupied by one state
- in the next layer: doubly occupied
- in the third layer: triply occupied etc. until a triangle layer is reached beyond which the multiplicity ceases to increase and remains $q+1$ for $p>q$ (or $p+1$ for $q>p$).
- it is easy to see that ψ_{max} has the third component of isospin:

$$t = t_3 = \frac{2\frac{p}{2}+q}{2} = \frac{2\frac{q}{2}+p}{2} = \frac{p+q}{2} \tag{78}$$



Representation $(p,q)=(5,1)$, Multiplicity of states at each site is indicated by the crosses.

How to show that finite representations are given by convex polygons?



Dimension of the representation

- The highest layer (triangle) has a side of length

$$\left| p - 2\,\frac{q}{2} \right| \;\text{ or }\; \left| q - 2\,\frac{p}{2} \right| = |p - q| \tag{79}$$

- Suppose $p > q$ each site of the highest triangle contains $q + 1$ states hence

$$\begin{aligned} \triangle_{top} &= (q+1)\left((p-q+1) + (p-q) + (p-q-1) + \ldots + 1\right) \\ &= \frac{q+1}{2}(p-q+1)(p-q+2) \end{aligned} \tag{80}$$

- The next layer contains

$$3\left((p-q+1)+1\right)q \tag{81}$$

  states

- The total number of states reads

$$\begin{aligned} d(p,q) &= \frac{1}{2}(q+1)(p-q+1)(p-q+2) + 3\sum_{k=0}^{q}(q-k)(p-q+2k+2) \\ &= \frac{1}{2}(p+1)(q+1)(p+q+2) \end{aligned} \tag{82}$$

$$d(p,q) = \frac{1}{2}(q+1)(p-q+1)(p-q+2) + 3\sum_{k=0}^{q}(q-k)(p-q+2k+2) \quad (83)$$

$$\star = \sum_{k=0}^{q}(q-k)(p-q+2k+2) = \sum_{k=0}^{q} q(p-q+2k+2) - \sum_{k=0}^{q} k(p-q+2k+2) \quad (84)$$

$$* = (q+1)q(p-q+2) + 2q\frac{(q+1)q}{2} = q(q+1)(p+2) \quad (85)$$

$$** = 2\sum_{k=0}^{q} k^2 + (p-q+2)\sum_{k=0}^{q} k \quad (86)$$

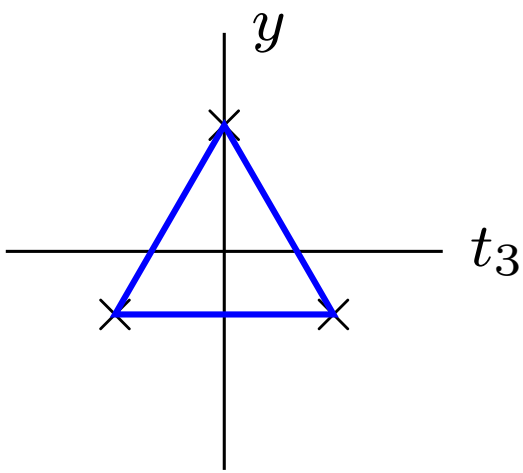
$$= 2\frac{1}{6}q(q+1)(2q+1) + (p-q+2)(q+1)\frac{q}{2} \quad (87)$$

$$= q(q+1)\left(\frac{2}{3}q + \frac{1}{3} + \frac{p-q}{2} + 1\right) = \frac{1}{6}q(q+1)(q+3p+8) \quad (88)$$

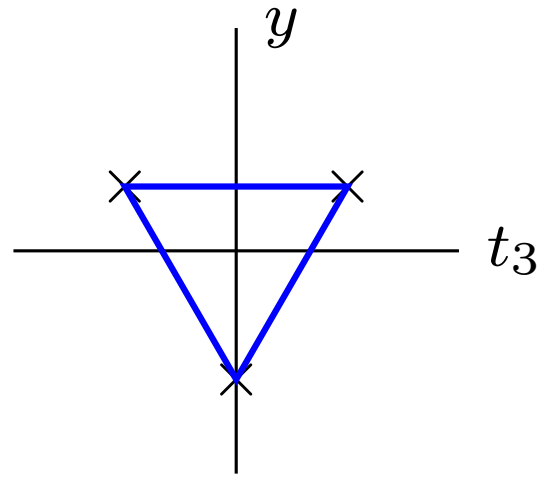
$$\star = q(q+1)(p+2) - \frac{1}{6}q(q+1)(q+3p+8) = \frac{q(q+1)}{6}(6p+12-(q+3p+8))$$

$$= \frac{q(q+1)}{6}(3p+4-q) \quad (89)$$

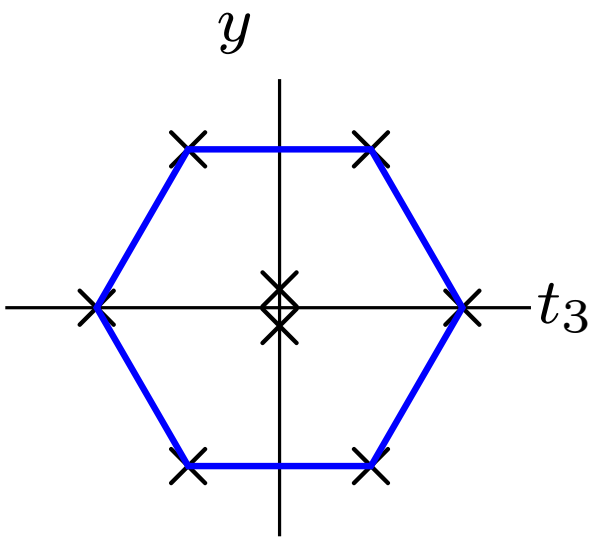
$$d(p,q) = \frac{1}{2}(q+1)(p-q+1)(p-q+2) + \frac{q(q+1)}{2}(3p+4-q) = \frac{1}{2}(p+1)(q+1)(p+q+2) \quad (90)$$



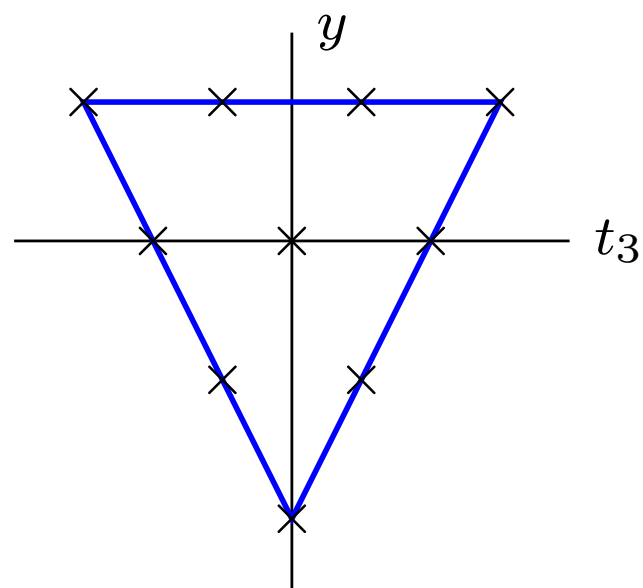
$(p,q) = (0,1) = \mathbf{3}^*$(triplet)



$(p,q) = (1,0) = \mathbf{3}$ (triplet)

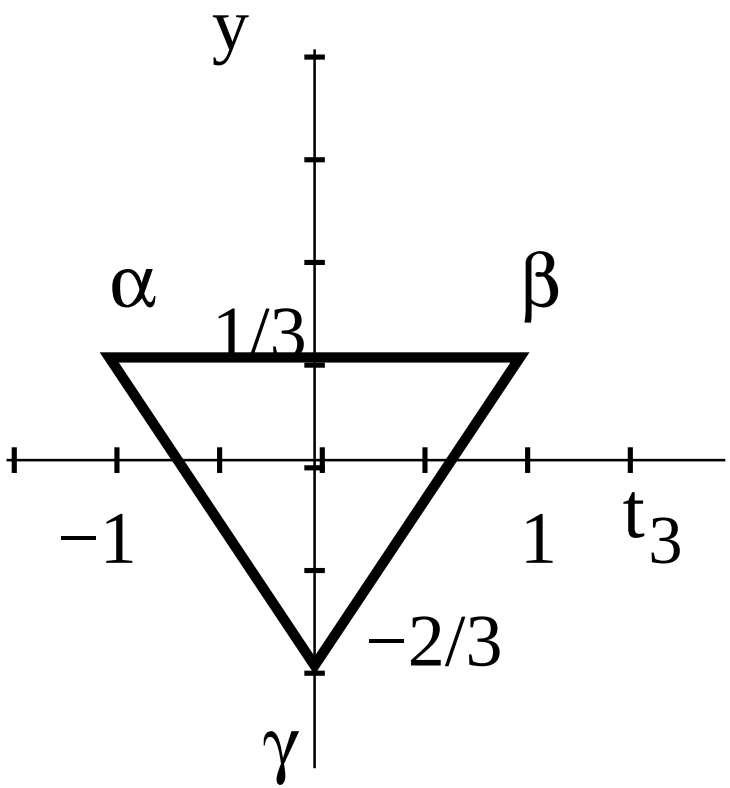


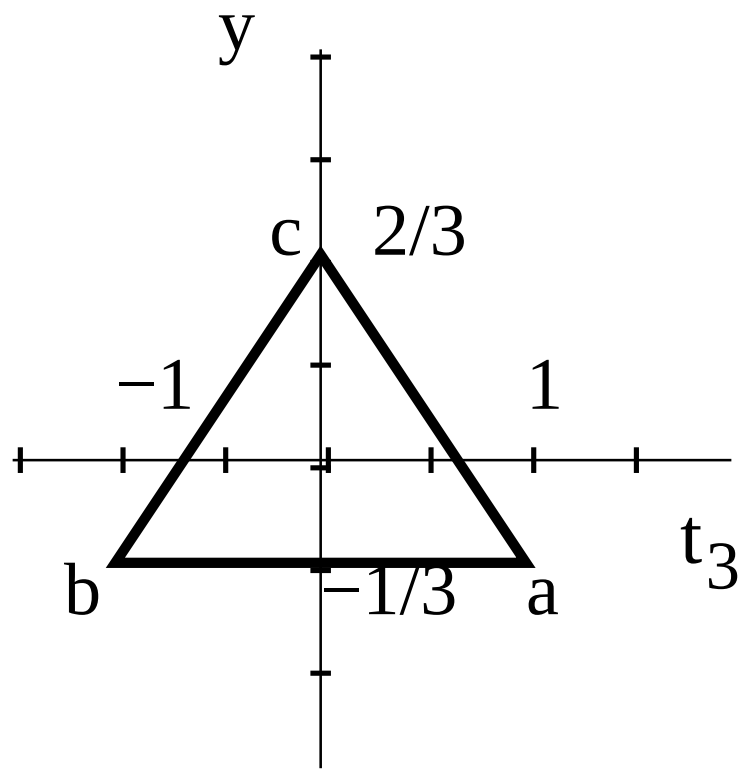
$(p,q) = (1,1) = \mathbf{8}$ (octet)



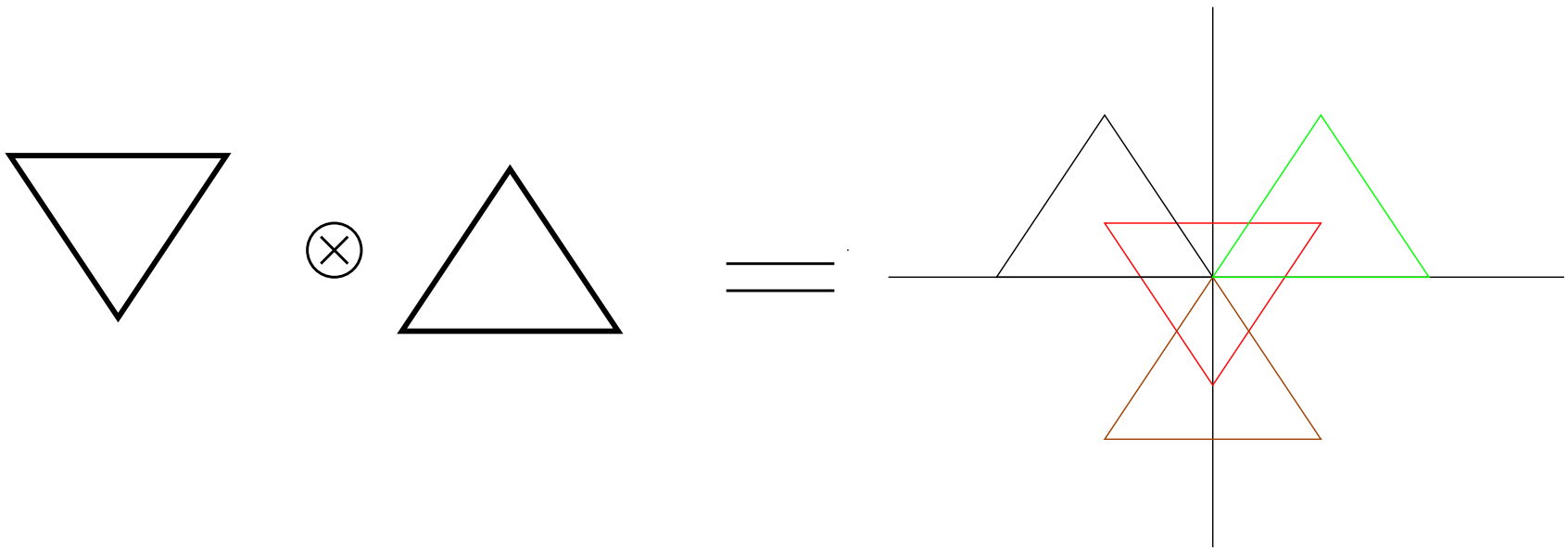
$(p,q) = (3,0) = \mathbf{10}$(decuplet)

Consider the tensor product $\mathbf{3} \otimes \mathbf{3}^*$

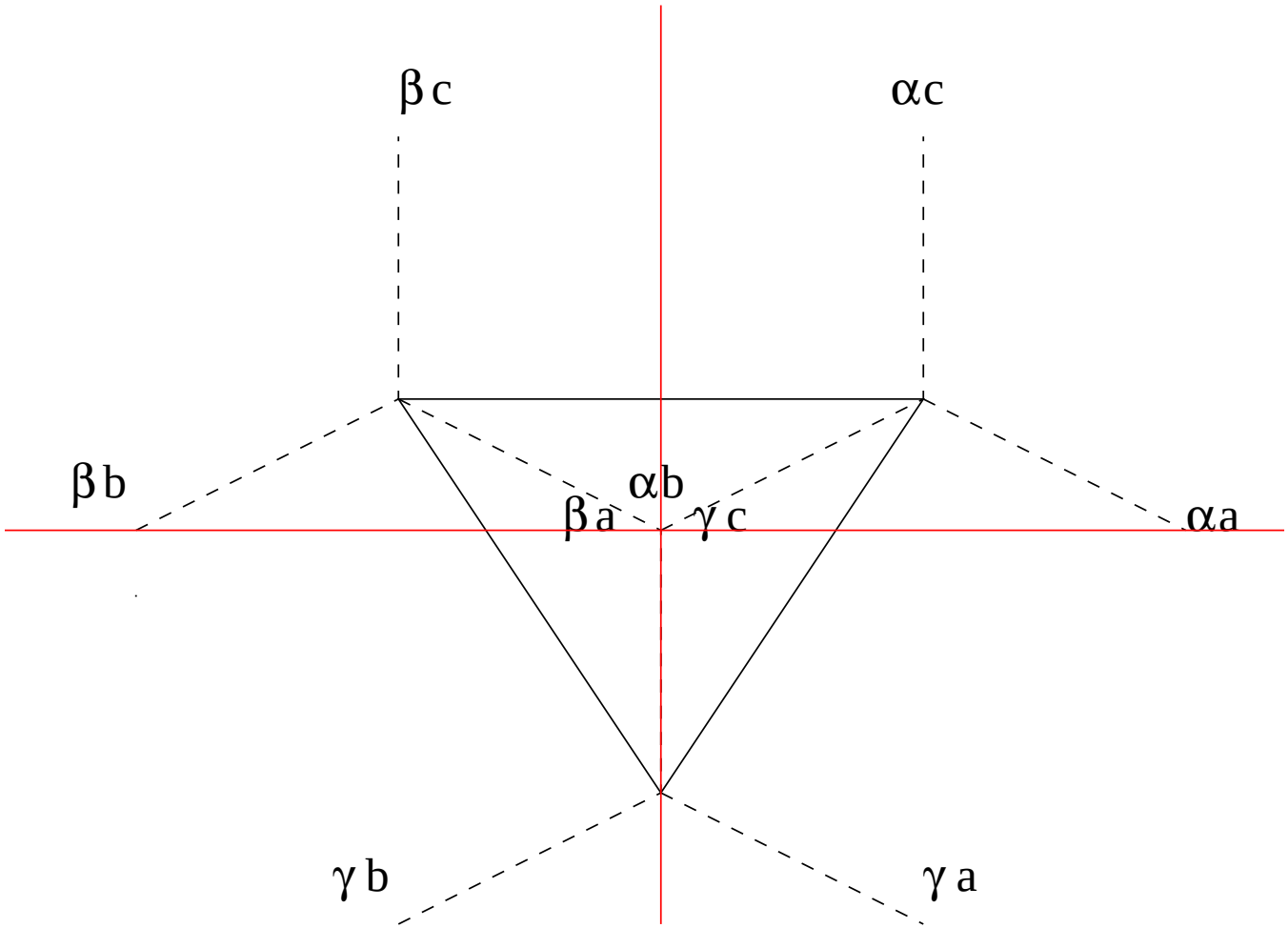


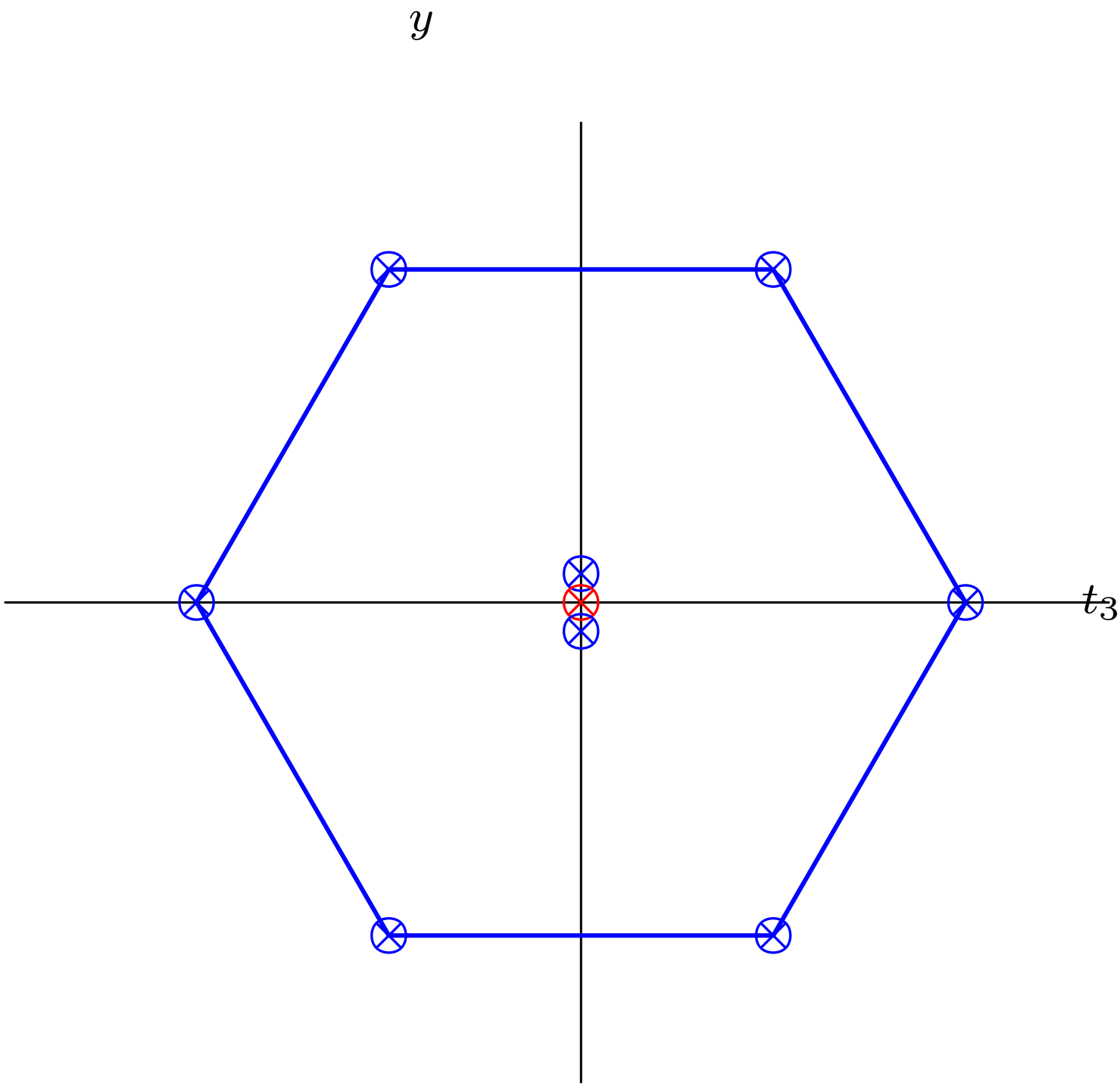


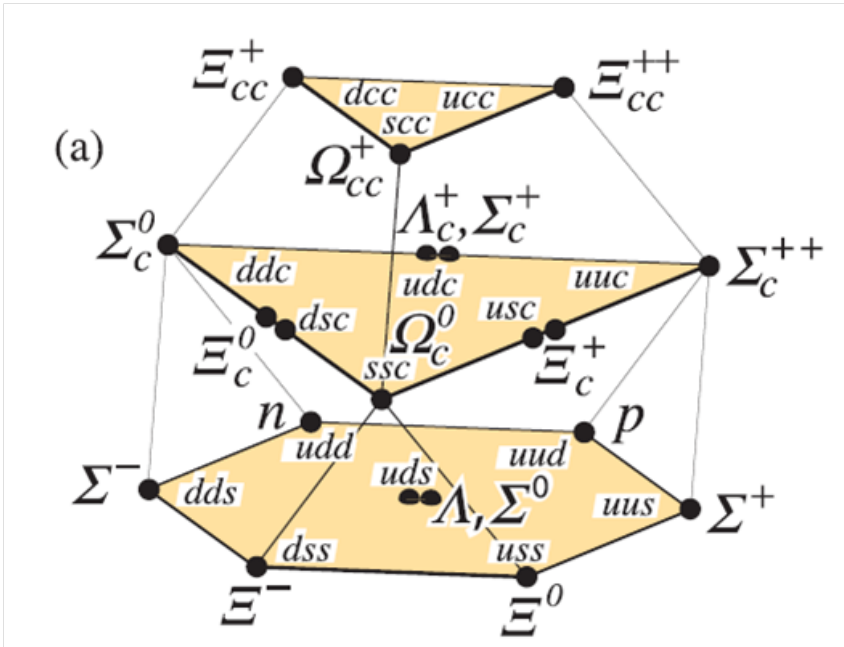
Consider the tensor product $\mathbf{3} \otimes \mathbf{3}^*$



Consider the tensor product $\mathbf{3} \otimes \mathbf{3}^*$




y
t_3



$SU(4)$ baryon multiplet, Fig. taken from PDG

- $SU(2)$ rank 1 group: 1-dimensional pictorial representations;
- $SU(3)$ rank 2 group: 2-dimensional pictorial representations;
- $SU(4)$ rank 3 group: 3-dimensional pictorial representations;
- $SU(n)$ ...

SU(n): tensor approach

- $$\psi_i = \begin{pmatrix} \psi_1 \\ \psi_2 \\ . \\ . \\ . \\ \psi_n \end{pmatrix} \tag{93}$$

  in space $\mathbb{C}^n$

  $$\psi'_i = U_{ij}\psi_j, \quad \psi'^*_i = U^*_{ij}\psi^*_j \quad U_{ij} \in SU(n) \tag{94}$$

- Scalar product

  $$(\psi, \phi) = \psi^*_i \phi_i \tag{95}$$

  is invariant to $SU(n)$ transformations

- $$\psi^*_i \equiv \psi^i, \quad U_i{}^j \equiv U_{ij}, \quad U^i{}_j \equiv U^*_{ij} \tag{96}$$

- Notice that

  $$\delta_{ik} = U_{ij}U^\dagger_{jk} = U_i{}^j U^*_{kj} = U_i{}^j U^k{}_j = \delta^k_i \tag{97}$$

- ψ_js are the basis for $SU(n)$ defining representations, also called fundamental or vector representation and denoted by $\mathbf{n}$
- ψ^js are the basis for the conjugate representation denoted by $\mathbf{n}^*$
- the basis of tensor of rank (q, p)

$$\psi'^{i_1 i_2 \ldots i_p}_{j_1 j_2 \ldots j_q} = (U_{j_1}{}^{k_1} U_{j_2}{}^{k_2} \ldots U_{j_q}{}^{k_q})(U^{i_1}{}_{l_1} U^{i_2}{}_{l_2} \ldots U^{i_p}{}_{l_p}) \psi^{l_1 l_2 \ldots l_p}_{k_1 k_2 \ldots k_q} \qquad (98)$$

- Kronecker delta is invariant under $SU(n)$ transformations

$$U_m{}^n U^j{}_k \delta^k_n = \delta^j_m$$

- Levi-Civita tensor is invariant under $SU(n)$ transformations

$$(U_{j_1}{}^{i_1} ... U_{j_n}{}^{i_n}) \epsilon_{i_1 ... i_n} = \det U \epsilon_{j_1 ... j_n} = \epsilon_{j_1 ... j_n} \tag{99}$$

We keep normalization

$$\epsilon^{j_1 \cdots j_n} = \epsilon_{j_1 \cdots j_n}$$

- Notice that:

$$\psi^i = \epsilon^{ii_2\ldots i_n}\psi_{i_2\ldots i_n}$$

Indeed

$$\psi'^i = U^i{}_k\psi^k$$

On the other hand

$$\begin{aligned}
\psi'^i = \epsilon^{ii_2\ldots i_n}\psi'_{i_2\ldots i_n} &= \epsilon^{ii_2\ldots i_n}U_{i_2}{}^{i'_2}\ldots U_{i_n}{}^{i'_n}\psi_{i'_2\ldots i'_n} && (100)\\
&= \epsilon^{ki_2\ldots i_n}\delta^i_k U_{i_2}{}^{i'_2}U_{i_3}{}^{i'_3}\ldots U_{i_n}{}^{i'_n}\psi_{i'_2\ldots i'_n} && (101)\\
&= \epsilon^{ki_2\ldots i_n}U_k{}^j U^i{}_j U_{i_2}{}^{i'_2}U_{i_3}{}^{i'_3}\ldots U_{i_n}{}^{i'_n}\psi_{i'_2\ldots i'_n} && (102)\\
&= U^i{}_j \underbrace{\epsilon^{ki_2\ldots i_n}U_k{}^j U_{i_2}{}^{i'_2}U_{i_3}{}^{i'_3}\ldots U_{i_n}{}^{i'_n}}_{\epsilon^{ji'_2\ldots i'_n}}\psi_{i'_2\ldots i'_n} && (103)\\
&= U^i{}_j\epsilon^{ji'_2\ldots i'_n}\psi_{i'_2\ldots i'_n} && (104)
\end{aligned}$$

- we always can limit ourselves to covariant tensors i.e. $\psi_{i_1 i_2\ldots}$

- Permutation of upper (or lower) indexes of tensor commutes with the group transformation
- Let P_{12} interchanges first with second indexes

$$P_{12}\psi'_{ij} = \psi'_{ji}$$

on the other hand

$$\psi'_{ji} = U_j{}^k U_i{}^l \psi_{kl} = \underbrace{U_j{}^l U_i{}^k \psi_{lk}}_{l \leftrightarrow k} = U_i{}^k U_j{}^l P_{12} \psi_{kl}$$

P_{12} commutes with group transformation.

- $SU(n)$ transformations do not change the symmetry properties of the tensor indexes!

* Tensors are bases for reducible representation of $SU(n)$!

- Let find the eigenstates of P_{12} operator

$$S_{ij} = \frac{1}{2}\left(\psi_{ij} + \psi_{ji}\right) \quad A_{ij} = \frac{1}{2}\left(\psi_{ij} - \psi_{ji}\right)$$

-

$$P_{12}S_{ij} = S_{ij}, \quad P_{12}A_{ij} = -A_{ij}$$

- S_{ij} and A_{ij} do not mix under the group transformation!

irreducible representations $\Rightarrow$ symmetry property

The task of finding irreducible representations of any arbitrary rank f (i.e. number of lower indexes) involves forming a complete set of permutation operations of these indexes.

Young Tableaux

- Examples of Young symmetrizers:

$$\begin{array}{|c|c|}\hline i & j \\ \hline \end{array} = \frac{1}{2}\left(\psi_{ij} + \psi_{ji}\right)$$

$$\begin{array}{|c|}\hline i \\ \hline j \\ \hline \end{array} = \frac{1}{2}\left(\psi_{ij} - \psi_{ji}\right)$$

$$\begin{array}{|c|c|}\hline i & j \\ \hline k \\ \cline{1-1} \end{array} = \psi_{ijk} + \psi_{jik} - \psi_{jki} - \psi_{kji}$$

- Indices appearing in the same row of the tableau are first subject to symmetrization.
- Subsequent indices appearing in the same column are subject to antisymmetrization.

A tableau where the index numbers do not decrease when going from left to right in row and always increase from top to bottom is a standard tableau

- Example:

$$\begin{array}{|c|c|}\hline i & j \\ \hline k \\ \cline{1-1}\end{array}$$

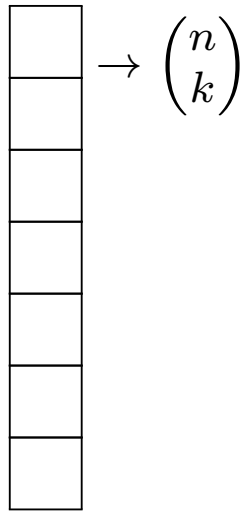
- Standard tableau

$$\begin{array}{|c|c|}\hline 1 & 1 \\ \hline 2 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1 & 1 \\ \hline 3 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 2 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 3 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 2 & 2 \\ \hline 3 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 2 & 3 \\ \hline 3 \\ \cline{1-1}\end{array}$$

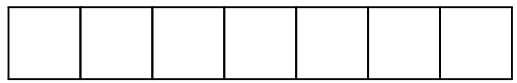
- Non-standard tableau

$$\begin{array}{|c|c|}\hline 1 & 1 \\ \hline 1 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 \\ \cline{1-1}\end{array},\ \begin{array}{|c|c|}\hline 2 & 1 \\ \hline 3 \\ \cline{1-1}\end{array}$$

  either vanish or are the linear combination of standard tableau

- Number of standard Young tableaux
- 

  Number of k-element subsets in $\{1, 2, ..., n\}$ set (order does not matter)
- Because of the anti-symmetrization, columns can not be longer than n; a column with n boxes can be contracted with $\epsilon_{1...n}$ and removed, hence at most $n-1$ rows remain

-  $\rightarrow \binom{n+k-1}{k}$

- it must be non-decreasing series of numbers
- to the i-th box entry we add $i-1$; we then obtain a k-element subset of $\{1,\ldots,n+k-1\}$ (without repetitions)

1	1	1	1	1	$\rightarrow$	1	2	3	4	5

1	1	2	3	3	$\rightarrow$	1	2	4	6	7

Fundamental theorem

A tensor corresponding to the Young tableau of a given patterns forms the basis of an irreducible representation of $SU(n)$. Moreover if we enumerate all possible Young tableaux under the restriction that there should be no more than $n-1$ rows, the corresponding tensors form a complete set, in the sense that all finite-dimensional irreducible representation of the group are counted only once.

dimensionality of irreducible representations

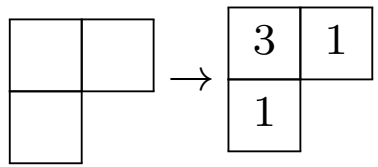
- if the Young tableau is characterized by the length of its rows $(f_1, f_2, ..., f_{n-1})$
- define the length differences of adjacent rows as

$$\lambda_1 = f_1 - f_2,\ \lambda_2 = f_2 - f_3, ..., \lambda_{n-1} = f_{n-1}$$

- The dimension of an $SU(n)$ irreducible representation will then be the number of standard tableaux for a given pattern

$$\begin{aligned} d(\lambda_1, \lambda_2, ..., \lambda_{n-1}) &= (1+\lambda_1)(1+\lambda_2)...(1+\lambda_{n-1}) \\ &\times \left(1+\frac{\lambda_1+\lambda_2}{2}\right)\left(1+\frac{\lambda_2+\lambda_3}{2}\right)...\left(1+\frac{\lambda_{n-2}+\lambda_{n-1}}{2}\right) \\ &\times \left(1+\frac{\lambda_1+\lambda_2+\lambda_3}{3}\right)\left(1+\frac{\lambda_2+\lambda_3+\lambda_4}{3}\right)\dots \\ &\left(1+\frac{\lambda_{n-3}+\lambda_{n-2}+\lambda_{n-1}}{3}\right) \\ &\dots \\ &\times \left(1+\frac{\lambda_1+\lambda_2+...+\lambda_{n-1}}{n-1}\right) \end{aligned}$$

- **hook length** : draw two perpendicular lines, in the shape of the hook, going to the right and another going to the downward. The total number of boxes that this hook passes, including the originating box itself is the hook length h_i associated with ith box.



- **distance of the first box** D_i is defined to be the number of steps going from the box in the upper left-hand corner of the tableau (the first box) to the ith box with each step towards the right counted as $+1$ unit and each downward step as -1 unit. For example we have

0	1	2
-1	0	
-2		

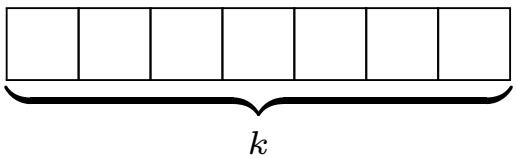
- The dimension of the $SU(n)$ irreducible representation associated with the Young tableau is given by

$$d = \prod_i \frac{(n + D_i)}{h_i}$$

- $SU(2)$

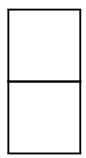
  $\square$: $d = (2+0)/1 = 2$ hence **2**

  $\square\square$: $d = ((2)/2) \cdot ((2+1)/1) = 3$ hence **3**

  $$d = \frac{2}{k} \cdot \frac{3}{k-1} \ldots \frac{2+k-2}{2} \cdot \frac{2+k-1}{1} = \frac{(k+1)!}{k!} = k+1$$

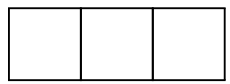
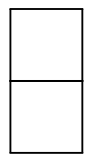
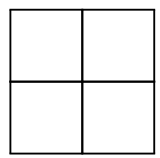
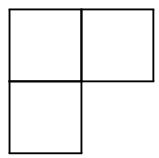
  so $\mathbf{k+1}$

- 

  $$d = \frac{2}{2} \cdot \frac{1}{1} = 1$$

  so **1** (singlet)

$SU(3)$

- $\psi_a =$  : **3** rank $(1,0)$
- $\psi_{ab} =$  : **6** rank $(2,0)$
- $\psi_{abc} =$  : **10** rank $(3,0)$
- $\psi^a = \epsilon^{abc}\psi_{bc} =$  : **3*** rank $(0,1)$
- $\psi^{ab} = \epsilon^{acd}\epsilon^{bef}\psi_{cdef} =$  : **6*** rank $(0,2)$
- $\psi^a{}_b = \epsilon^{acd}\psi_{cdb} =$  : **8** rank $(1,1)$